EUROPEAN UNION STEM LABORATORY INITIATIVE

SPHERE: MISSION CONTROL

STUDENT LABORATORY MANUAL & TASK SHEET

Inquiry-Based Science Education (IBSE) Framework: This manual guides you through building a physical thermal spacesuit capsule, capturing real-time temperature telemetry, comparing physical experimental data against a live computer model (Digital Twin), and plotting results on a manual laboratory task sheet.

1. INTRODUCTION & EXPERIMENTAL OBJECTIVES

Space is one of the most thermally hostile environments known to humanity. In Low Earth Orbit (LEO), spacecraft and extravehicular astronauts experience solar radiation heating exterior surfaces to over **+121°C** in direct sunlight, while dropping to **-157°C** when shielded in Earth's orbital shadow.

To keep astronauts alive, aerospace engineers design Multi-Layer Insulation (MLI) systems to regulate thermal energy transfer. In this experiment, your team will act as Mission Controllers. You will construct a physical spacesuit thermal envelope for a simulated astronaut (a 100ml core filled with hot water at 80°C) submerged in an ice-water chamber (0°C).

Using the **SPHERE: Mission Control** web dashboard, you will stream physical telemetry, compare live readings with Newton's Law of Cooling, and compute your crew's manual Correlation Score.

2. HARDWARE ASSEMBLY GUIDE & SCHEMATIC

The laboratory apparatus uses standardized, accessible components to create a waterproof thermal capsule capable of sustained ice-water submersion:

COMPONENT	SPECIFICATION	PURPOSE IN EXPERIMENT
PET Core Jar	100ml wide-mouth PET bottle	Simulates the core astronaut environment containing hot water.
Cable Gland	Industrial PG7 Nylon Gland	Mounted through the cap to create a 100% watertight wire entry.
Digital Sensor	DS18B20 Waterproof Probe	Measures water core temperature continuously inside the capsule.
Thermal Chamber	Ice Water Bath (0°C)	Simulates the extreme cold thermal sink of deep space.

Assembly Instructions:

1. Unscrew the dome cap of the PG7 gland and thread the DS18B20 temperature sensor probe through the gland body.
2. Insert the threaded PG7 gland through the 12.5mm pre-drilled hole in the PET jar cap and secure it firmly with the internal locknut.
3. Tighten the external PG7 dome nut until the neoprene rubber seal clamps snugly around the sensor cable. This prevents liquid intrusion during submersion.
4. Ensure the temperature sensor tip hangs suspended in the exact center of the 100ml jar without touching the container walls.

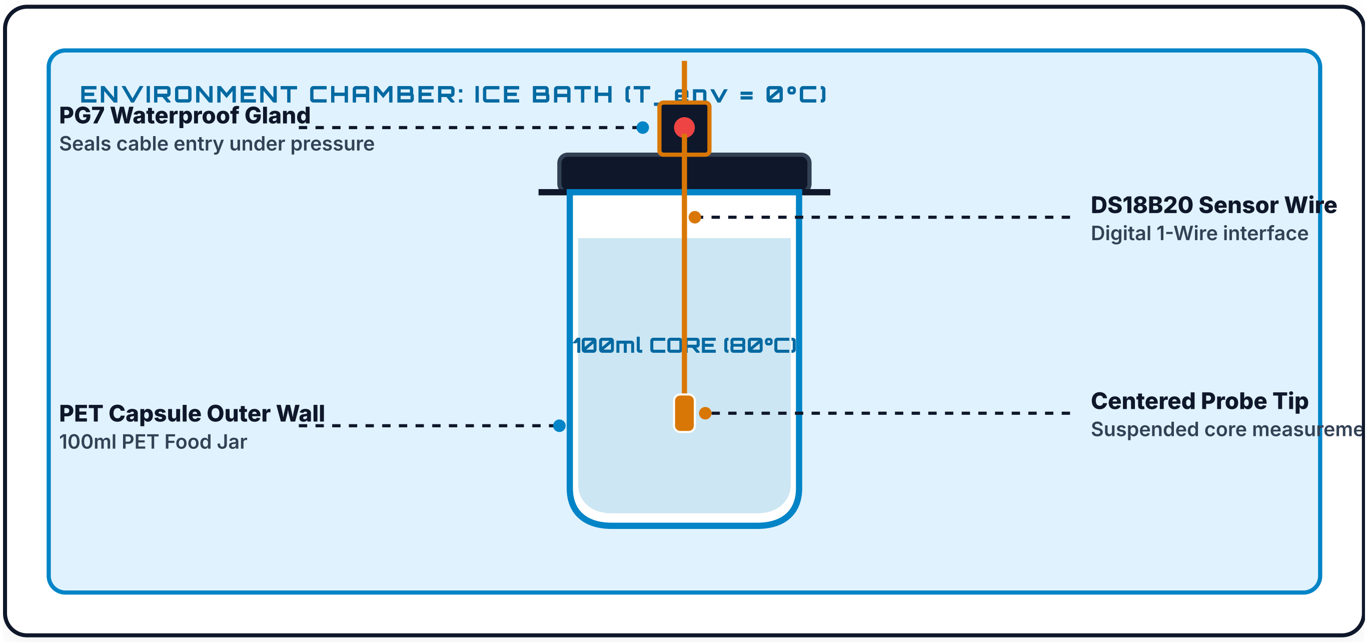


FIGURE 1: MECHANICAL ASSEMBLY DIAGRAM & THERMAL PROBE SEEDING

3. THERMAL PROTECTION SYSTEM (TPS) CONFIGURATIONS

Your engineering team will test one of four thermal protection configurations. Each system isolates different modes of heat transfer:

A. BARE CONTROL

`k = 0.150`

Uninsulated PET jar directly exposed to liquid. Tests baseline conductive & convective dissipation.

B. COTTON WRAP

`k = 0.080`

Single fabric barrier. Traps stationary air pockets to reduce conductive heat transfer.

C. MYLAR FOIL

`k = 0.040`

Reflective metallic film. Reflects radiant thermal electromagnetic energy back to core.

D. FULL MLI

`k = 0.015`

Multi-layer system combining Cotton + Bubble Wrap + Mylar for complete conduction/convection/radiation defense.

4. MATHEMATICAL MODELING & NEWTON'S LAW OF COOLING

Heat naturally flows from regions of higher temperature to lower temperature. The rate of cooling is proportional to the temperature differential between the object and its environment:

NEWTON'S LAW OF COOLING (EXPONENTIAL DECAY MODEL)

$$T(t) = T_{env} + (T_0 - T_{env}) \cdot e^{(-k \cdot t)}$$

Where: **T(t)** = Predicted Temp (°C) at minute *t* | **T_env** = Ice Bath Temp (0°C) | **T_0** = Initial Core Temp (80°C) | **k** = Cooling Rate Constant (1/min)

5. STEP-BY-STEP MISSION CONTROL WORKFLOW

1. **Pre-Flight Certification:** Access Mission Control on the desktop terminal and complete the Pre-Flight Safety Quiz to initialize telemetry access.
2. **Run Simulation Prediction:** Go to the **Simulator** tab. Select your assigned insulation configuration, set initial parameters ($T_0 = 80^\circ\text{C}$, $T_{\text{env}} = 0^\circ\text{C}$), and click **RUN SIMULATION PREDICTION**.
3. **Prepare Physical Water Core:** Measure 100ml of hot water at 80°C, pour it into the PET jar, screw the PG7 cap tightly, and verify the waterproof seal.
4. **Submerge & Start Timer:** Place the capsule into the 0°C ice water bath and click **START LIVE TIMER** in the **Live Telemetry** tab.
5. **Log Measurements Every 60 Seconds:** Record the thermometer reading into your Task Sheet and the Mission Control digital entry log at every minute mark up to 15 minutes.
6. **Compute Correlation Score:** Compare your paper graph curve against the Digital Twin overlay!

SECTION 6: STUDENT TASK SHEET & GRAPH WORKBOOK

WORKSHEET #01

CREW MEMBERS / NAMES:

____

CREW CALLSIGN / TEAM ID:

____

DATE / CLASS PERIOD:

____

TESTED INSULATION CONFIGURATION:

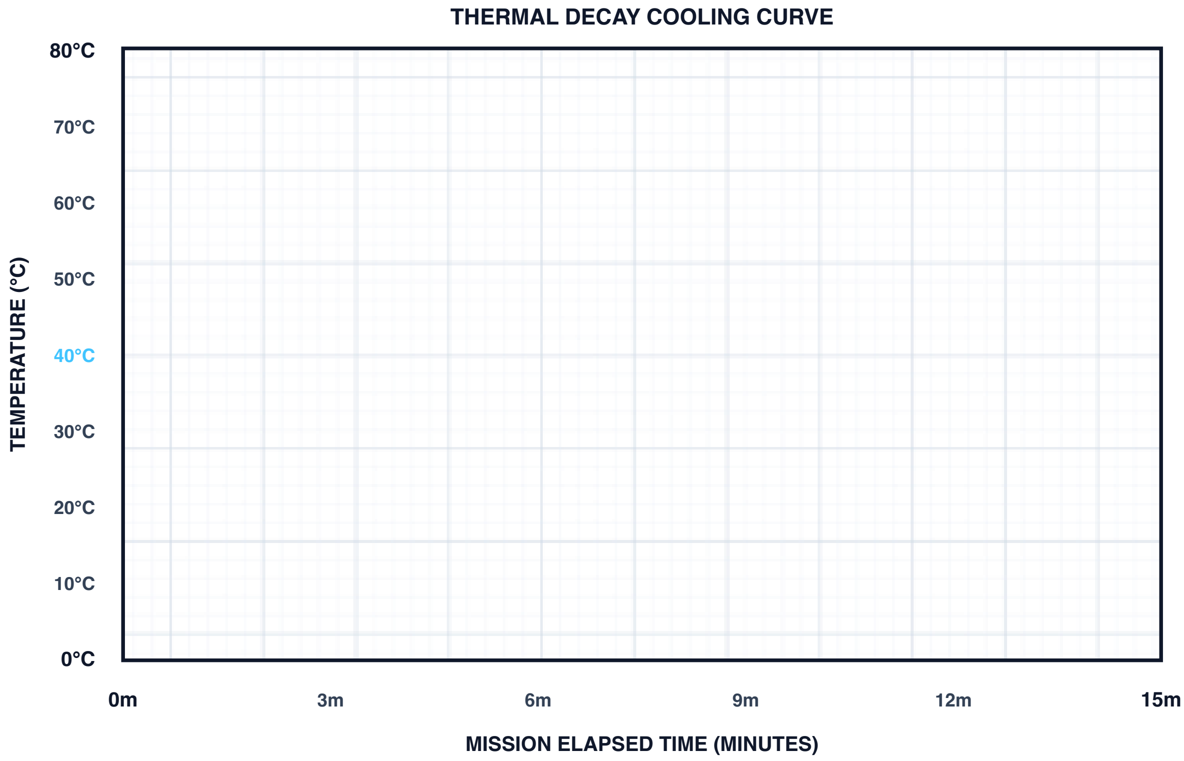
[ ] A: Bare   [ ] B: Cotton   [ ] C: Mylar   [ ] D: MLI

Part A. Live Telemetry Data Logging Table

TIME (MIN)	PHYSICAL TEMP (°C)	PREDICTED TEMP (°C)	ABS DIFFERENCE |ΔT|	OBSERVER FIELD NOTES
0 min	80.0°C	80.0°C	0.0°C	Capsule submerged in 0°C ice bath
1 min				
2 min				
3 min				
4 min				
5 min				
6 min				
7 min				
8 min				
9 min				
10 min				
11 min				
12 min				
13 min				
14 min				
15 min				Final measurement checkpoint

Part B. Printable Coordinate Graph Grid

Plot physical measurements as points **(•)** and connect with a solid line. Draw theoretical predicted temperatures as a dashed curve **(---)**.



Part C. Manual Thermal Correlation Score Calculation

Calculation Steps:

1. Sum the absolute differences in Column 4: **Sum |ΔT| = ____________ °C**
2. Calculate Average Error: **Avg Error = Sum / 15 = ____________ °C**
3. Compute Correlation Score: **Score = 100 - (Avg Error × 5) = ____________ %**

FINAL CREW PERFORMANCE RATING: [ ] A+ (95%+) [ ] A (90%+) [ ] B (80%+) [ ] C (70%+) [ ] RE-CALIBRATE (<70%)

7. LAB ANALYSIS & REFLECTION QUESTIONS

Question 1: Thermal Radiation vs Conduction

Mylar is designed to reflect infrared radiation in space. Why does Mylar alone perform poorly when placed directly in an ice-water bath without an inner cotton or bubble layer?

Question 2: Physical Meaning of Cooling Constant (k)

In Newton's Law of Cooling, does a smaller k value represent a better thermal insulator or a worse thermal insulator? Explain using the mathematical exponential formula.

Question 3: Multi-Layer Insulation (MLI) Synergy

Explain how combining Cotton + Bubble Wrap + Mylar in Configuration D creates a synergistic effect that outperforms any individual insulating material.

Question 4: Physical vs Digital Twin Discrepancy

Identify two potential sources of experimental error that could cause your physical sensor readings to deviate from the computer model prediction.

Question 5: Environmental Boundary Adjustments

If your classroom ice bath warms up from 0°C to 5°C during the experiment, how should you adjust the simulation inputs in Mission Control to preserve model validity?

Question 6: Water Volume & Heat Capacity Scaling

If the water volume inside the PET capsule was increased from 100ml to 200ml, how would the total heat capacity ($Q = m \cdot c \cdot \Delta T$) affect the cooling rate? Would k increase or decrease?

Question 7: Spacesuit Design Trade-Offs in EVA

In real space missions (such as spacewalk Extravehicular Activities), why can't engineers simply add 50 layers of insulation to achieve a near-zero k value? What mechanical trade-offs occur?

Question 8: Correlation Score Optimization

Describe how your crew can refine your measurement techniques or hardware sealing to improve your Thermal Correlation Score from 75% to 95%+.